जय माँ सरस्वती

UNIT-3

NOISE POLLUTION

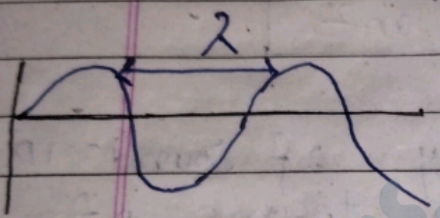
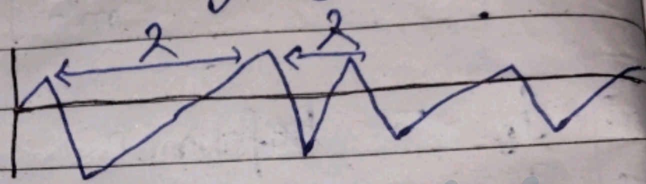
★ Noise pollution:-

- Noise pollution is unwanted or excessive Sound that can have harmful effect on human health and enviromental Quality.
- It is also known as enviromental noise or sound pollution.

- Noise pollution also caused when the Loudness of Sound become Irritating or Unbearable.

★ Sources of noise pollution:-

- Transport Vechicles
- Domestic gadget
- personal entertaiments
- functions
- Cracker and detonations
- Loud Speakers
- Comercial establisment
- Industries

Sound	Noise
① It is pleasant to hear	① It is Unpleant to hear.
② Sound wave has periodic motion.	② Noise wave has non periodic motion.
③ pitch of wave is Constant.	③ pitch of wave is changing.
	
④ It produce meaningful Communication	④ It produce non meaningful communication
⑤ It's unit is Hz	⑤ Its unit is (dB) decibel.

* Noise pollution :—
- Sound waves are vibration of air molecules carried from noise sources to ear.
- The unit of Sound Intensity is decibel (dB)
- Human can hear a conversation at 60dB.
- A/q to the world Health organization; sound level less than 70 dB are not damaging to living organism, regardless of how long or consistent the exposure is.

- Exposure for more than 8hrs to constant noise beyond 85dB may be hazardous
- Sound Intensity of 130dB is upper limit of the threshold of hearing and beyond this it may damage ear.

* Sources of noise pollution :—

- The sources of noise pollution can be broadly divide into :
1) Natural Sources
2) Man-made Sources

* Natural Source :-
- The rains, winds and thunders fall under this category.
- They are occasional occurrence.
- Sometimes they are very damaging.

* Man - made Sources :-
It is divided into two Categories.
a.) Indoor Sources
b.) Outdoor Sources

(a) Indoor Sources :-
- Noise pollution inside the ~~house~~ home from the electrical and electronic gadgets used in daily life.

Ex→ T.V, Audio, grinder, pressure Cooker, vaccum cleaner, dryers, Coolers, Air Conditioners and washing machine.

(b) Outdoor Sources :-
(i) Industrial Sources
Various types of noise like hammering sounds, whistling sounds crushing sounds, vibrating sounds are emitted from the Industries

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· mining Activites :-
It Include blasting of rocks, drilling of rocks, underground mining Activities etc.

(b) Outdoor Sources :-
(ii) Non - Industrial Sources :-
· Traffic
  · Road Traffic
  · Rail Traffic
  · Air Traffic

* Construction Activities
· Noise of machines tools and loud noise of workers on site.

· Social Events
- Marriages religious functions, parties and events having loud speaker

* festive occasions
· Drum beats, loud speaker, prayers and cracker.
* power generation
* Agriculture equipment

* Types of noise :-
(i) Impulsive noise
(ii) Continous noise
(iii) Intermittent noise

(i) Impulsive noise :-
- It is noise characterized by a noise level than 40dB within half second with a duration of less than one second.
- It includes unwanted almost instantaneous sharp sound.

Eg :- firing of weapon, explosions or drop forge impact etc

(ii) Continous noise :-
- The noise which produce continously for example by machinery that keeps running without interruption.

(iii) Intermittent noise :-

- The noise that is not produced continuously but with breaks.
Example - Drilling machine used by carpenter or a dentist.

* Effect of noise pollution on Health

- Noise pollution impacts millions of people on a daily basis. The most common health problem it cause is Noise Induced Hearing Loss (NIHL)

- Exposure to loud noise can also cause high blood pressure, heart disease, sleep disturbances and stress.

- These health problems can effect all age groups, especially children.

- Many children who live near noisy airports or streets have been found to suffer from stress and other problems, such as impairments in memory, attention level, and reading skill

* The various problems generated due to noise pollution can be studied under two head
1. Auditory problems
2. Non Auditory problems

* Auditory problems :-

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ⓐ Hearing problems :-
The continuous exposure of unwanted sound for long duration may result in the damage of ear drums and subsequently loss of hearing.

ⓑ Auditory fatigue :-
• By continuous exposure to noise of various intensities restlessness and tiredness occurs. It is known as auditory fatigue.

★ Non Auditory problems :-

ⓐ Health Issue :-
• The non-Auditory health issue triggered by noise pollution include the problems of cardiovascular system, nervous system, gastro-intestinal, headache, nausea & vomiting psychological disturbance irration etc.

ⓑ Sleeping Disorder :-

• The noise of various intensities may disturb the sleep.

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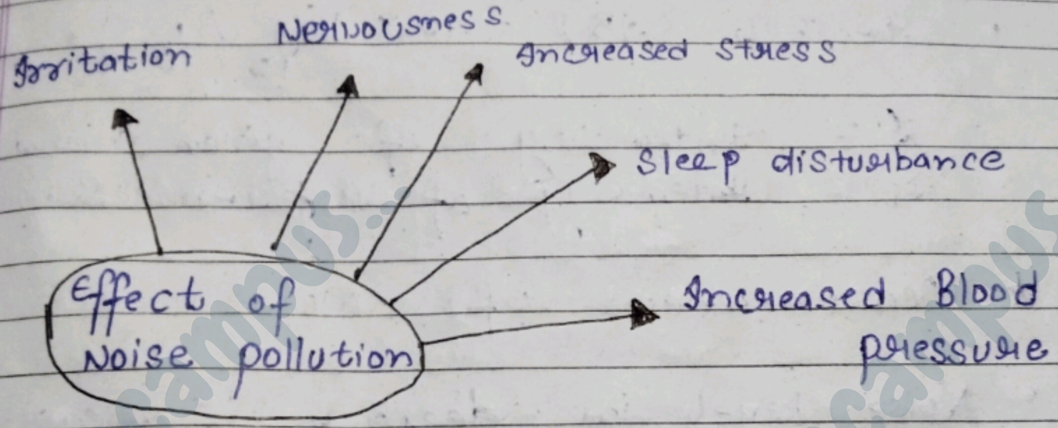
- The disturbance may include shorter sleep duration, frequently awaking and development of anxiety and irritation take place.



(c) Interference in communication :—

- Noise pollution hinders the communication process in several ways between source and receiver.

(d) Effect on wildlife

- Noise has large impact on wildlife.
- Generally animals easily get disturbed with noise.
- That's why in prohibited area like national parks and sanctuaries the use of horns is prohibited.

★ Measurement of noise :—

- The Intensity of Sound is measured in terms of Sound pressure level and common unit is decibel.

- $\text{Decibel (dB)} = 10 \times \log_{10}\left(\frac{I}{I_0}\right)$

  I = Intensity of given Sound
  I_0 = Intensity of reference Sound

★ Measurement of Noise :—

- Noise level in decibel is measured with an instrument called Sound Level meter (SLM)

- It is also known as noise meter.

- SLM is a digital measuring instrument
- It is placed at the place where measurement of noise required.
- The Instrument must Always be positioned away from direct Source vibration and from obstruction.

- After placing it. it gives reading on screen.
- Once we get the reading then we can compare them with standards to understand the level of noise.

★ Sources of noise pollution :—

① Traffic Sources :-
- Horn of vechicles.
- Raise of accelerator.
- Vechicle with damage Silancer.
- Noise produced by disel car is more than petrol car.
- A jet air craft produces more noise than a propeller type air craft.

② Industrial sources :-
- Reciprocating & Rotating machineics.
- Grainding wheals
- Blow hammers
- Generators
- High pressure and High velocity gases

③ Constructional Sources :—
- Rock Crusher
- pile driving equipments.
- Boaring & drilling devices.
- Road Roller
- Belt conveyers
- Rock blasting.

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④ Other Sources :-
Residental area — loud voice of T.V., music system, radio etc.
public places :- public functions, Navratri festival.
Siren :- police van, Ambulance & Industrial use of crakers during diwali shorting of hawkers of market playing of children at play ground.

* Effect of noise pollution:
① physical effect
- Loss of hearing - Long time exposure of loud (80-90) dB) for more than 8hrs a day may cause Loss in hearing.
- Total deafness (Acoustic trauma).
- Sudden Loudness Like crakers & blast may effect the ear drum.

② ~~Physical~~ physiological effects :
- Headache nausea.
- Gastric ulcers & Acidity.
- High rate of heart beat.
- ~~for~~ fluctuation in blood pressure and Sugar Level.

③ Effect on Animals :-

- Hearing loss result from noise level 85dB or more.
- masking Ability to hear environment signals.
- Increased heart rate and respiration.
- Behavior effect.

④ Effect on plants:
- Due to noise pollution of flowers & their spreading seeds are affected.
- Noise pollution effect on plants remains last long even after source of the noise goes away.

* Control of noise pollution:-
① Control of receiver's End
- people working in noisy area car-protection aid like - ears plugs, ear-muffs, noise helmates and head-phones must be provided.

② Supression of noise at source.

- Installing noisy machines in sound proof chamber.
- proper maintance and lubrication of machines.
- Using silencers to control noise from Automobile.

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- Designing, fabricating and using quiter machines.
- Reducing noise from vibrating machines.

③ Acoustic Zoning :-
- Silence zones nearer to educational hospital. and residencial area should require.
- Increasing distance between sources & reciver of noisy Industrial area, bus terminals, railway stations & aerodromes away from residential area.

④ planting Trees :-
- planting green trees along road side, near hospitals, school will reduce the noise pollution.

⑤ Sound Insulation of Construction stages :-
- Gaps between doors & wall should be packed with sound absorbing materials.
- In auditorium perforated plywood should be fixed on walls & ceilings.

* Intensity of Sound :-

- The Amount of Sound energy passing through unit area per Second in the direction perpendicular to area.

$\boxed{I = \text{unit } \frac{J/s}{m^2}}$ or $\boxed{\frac{\text{Watt}}{m^2}}$

- I_0 = Reference Intensity of Sound $= 10^{-2} w/m^2$
- Sound is measured in terms of decimal which is the ratio of I & I_0

$\boxed{dB = 10 \log_{10} \left(\frac{I}{I_0}\right)}$ (Pitch & frequency is not considered)

- Consideration of frequency and pitch is very Important. (Two diff^n Sound)
- Modified Scale to measure the Sound is dBA.

3. Noise pollution	Noise pollution: sources of pollution, Measurement of Noise pollution level.	CO3	1,5,7
	Effects and Control of Noise pollution. Noise pollution (Regulation and Control) Rules, 2000	CO3	1,5,7